INCH-POUND

MIL-V-87255 (USAF)
31 December 1990

VALVES, PRESSURE, ANTI-G SUIT, HIGH FLOW

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high-flow, pressure regulating valve for inflating anti-G suits and providing a sense line signal for a g-compensated oxygen regulator (if applicable). It also includes details of inlet and outlet fittings required to interface the valve to specific aircraft.

1.2 Classification. The designations for the three valves shall be as follows:

a. VAU-45(V)1/A
b. VAU-45(V)2/A
c. VAU-45(V)3/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-PattersonAFB OH 45433-6503, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
PPP-B-636	Boxes, Shipping, Fiberboard MILITARY

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Federal

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment

Military

DoD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-2073-1	DoD Material Procedures for Development and Application of Packaging Requirements
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal
MS33658	Fitting End, Hose Connection, Standard Dimensions For

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, ATTN: NPODS, 5801 Tabor Avenue, Philadelphia PA 19120-5099.)

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2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials. Materials shall conform to applicable specifications and standards as specified herein and on applicable drawings. Materials which are not covered by specifications and standards, or specifically described herein, shall be reported to the acquisition activity and shall require contracting officer approval prior to use. Materials shall be of the best quality, of the lightest practical weight, and suitable for the purpose intended.

3.3.1 Metal parts. All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.3.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.3.2 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with air shall not be used.

3.3.2.1 Elastomer components. Elastomer components, except silicone, shall be not more than 12 months old from the date of manufacture to the date of delivery to any government service or to any airframe or accessory manufacturer.

3.3.3 Protective treatment. Materials shall be protected from deterioration in a manner that shall in no way prevent compliance with the performance requirements of this specification. The valve shall resist corrosion and deterioration due to fuels, salt fog, microorganisms, moisture, hydraulic fluid, lubricating oils, grease, ozone, or atmospheric conditions likely to be encountered in storage or normal service. Protective coatings which might crack, chip, or scale during normal service or storage life or under extremes of environmental conditions shall not be used.

3.4 Standard parts. Military standard parts shall be used when they suit the purpose. Use of non-standard parts requires contracting officer approval prior to use.

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3.5 Design and construction. The valve shall be designed to pass air at a regulated pressure to the aircrew member's anti-G suit and to the sense line of the regulator (if applicable) and shall contain provisions for suit exhaust. The valve shall include provision for inlet and outlet fittings that incorporate swivel features. The fittings shall have a diameter and finish compatible with the aircraft inlet hose, anti-g suit delivery hose, and the outlet pressure sense line (if applicable). The outlet fitting shall be in accordance with Figure 1, 2 or 3, as required. The outlet pressure shall be a function of the acceleration force applied to the valve in the +Gz (see 6.4.1) direction. The design shall enable manual operation (press-to-test) of the valve and shall provide means to simulate acceleration forces (see 3.9.2 and 4.3.2). The valve shall be designed to perform as specified in 3.6. Moisture in the air source to the valve shall not adversely affect operation of the valve.

3.5.1 Relief valve. The valve shall include a pressure relief valve that shall not allow the valve outlet pressure to the anti-g suit and the sense line (if applicable) to exceed 13 psig.

3.5.2 Lubrication. The valve shall be designed and constructed in such a manner that no maintenance lubrication shall be required.

3.5.3 Inlet screen. The valve shall contain a corrosion-resistant steel, 30 by 30 maximum mesh, 0.010 inch minimum wire diameter inlet screen. All inlet air shall pass through the screen.

3.6 Performance. All specified performance requirements shall be based on the use of air.

3.6.1 Minimum operating acceleration force. The minimum equivalent acceleration force required to open the valve to permit flow of 0.01 liter per minute shall be between 1.25 and 1.50 Gz (see 4.6.2).

3.6.2 Outlet pressure regulation. For a constant applied acceleration force, the outlet pressure schedule during testing (see 4.6.3) shall be within the range specified in Table I and displayed graphically in Figure 4 for both gradual onset/offset (<= 0.1 Gz/sec) and rapid onset/offset (= 4.0 Gz/sec). Outlet pressure shall be met within 2.0 seconds of reaching the constant force when accelerating and within 5.0 seconds of reaching the constant force when decelerating. The outlet pressure shall not fluctuate more than $\pm$0.1 psi at any given constant acceleration force.

3.6.3 Leakage. The leakage to the ambient shall not exceed 1.0 liters per minute. There shall be no leakage to the outlet (see 4.6.4).

3.6.4 Anti-chatter. The valve shall not chatter (see 4.6.5 and 6.4.2).

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Table I. Outlet pressure.

G-Range Gz	Maximum Pressure PSIG	Minimum Pressure PSIG
0 - 1.80	-	0
1.80 - 2.00	-	Gz - 1.80
2.00 - 4.00		1.40*Gz - 2.60
4.00 - 9.33	-	1.50*Gz - 3.00
> 9.33		11.00
1.25 - 4.00	1.45*Gz - 1.80	-
4.00 - 10.00	1.50*Gz - 2.00	-
> 10.00	13.00	-

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3.6.5 Outlet pressure response

3.6.5.1 Response Time. When the acceleration force is being applied by a centrifuge or an aircraft, the outlet pressure shall be in accordance with Table I (Figure 4) within 2 seconds after the application of the force. When the acceleration force is being decreased, the pressure shall be in accordance with Table I (Figure 4) within 5 seconds after decreasing the force. Additionally, the outlet pressure shall change in a smooth continuous manner without oscillations or step responses.

3.6.5.2 Test Limit Response Time. During testing using a step response input to simulate the G force the tank pressure shall have the response time given by Table II (see 4.6.6).

3.6.6 High temperature exposure. The valve shall operate satisfactorily after completion of the high temperature test at 250°F ±5°F (see 4.6.7).

3.6.7 Low temperature exposure. The valve shall operate satisfactorily after completion of the low temperature test at -65°F ±5°F (see 4.6.8).

3.6.8 Vibration. The valve shall operate satisfactorily and shall show no evidence of mechanical or material failure after completion of the vibration tests (see 4.6.9).

3.6.9 Salt fog. The valve shall show no evidence of corrosion of mechanical failure after completion of the salt fog test (see 4.6.10).

3.6.10 Internal corrosion. The valve shall show no evidence of corrosion of mechanical failure after completion of the salt fog test (see 4.6.11).

3.6.11 Centrifuge testing. This testing shall be done using valves of the same design as those that passed all other performance tests in section 3.6. Centrifuge testing shall include an anti-chatter test and performance testing (see 4.6.12).

3.6.12 Endurance. The valve shall have a mean-cycle-between-failure of 15,030 cycles (see 6.4.3) and shall pass the endurance test (see 4.6.13).

3.6.13 Disassembly and examination. The valve shall not show excessive evidence of deterioration or wear due to testing which might affect the valve performance (see 4.6.14).

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with each other with respect to installation and performance. The item identification and part number requirements of DoD-STD-100 shall apply.

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Table II. Test Response Time.

Force Applied From	Tank Pressure	Response Time (maximum)
1 to 10 Gz	As specified on Figure 4 for 10 Gz	1.6 sec
10 to 1 Gz	0	2.7 sec
1 to 6 Gz	As specified on Figure 4 for 6 Gz	1.6 sec
6 to 1 Gz	0	2.2 sec

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3.8 Weight. The total weight of the valve shall not exceed 3 pounds.

3.9 Identification of product. The valve shall be marked for identification in accordance with MIL-STD-130 and shall include at least the following information:

Valve, Pressure, Anti-G Suit,
1/

Manufacturer's Part No.

Manufacturer's Identification

US MXXXXX-X (see 6.5)

Date of Manufacture: MFD (Month)(Year) 2/

1/ Enter the applicable designation (VAU-45(V)1/A, VAU-45(V)2/A, or VAU-45(V)3/A) here.

2/ The manufactured date may be stamped on the valve rather than marked on the nameplate and shall be as permanent as the normal life expectancy of the valve.

3.9.1 Ports. All inlet and outlet ports shall be clearly identified by permanent markings.

3.9.2 Press-to-test button. The valve shall have a permanent marking on the top of the press-to-test button to identify it as a "HI FLOW" for the VAU-45(V)1/A and as "HF PBG" for VAU-45(V)2/A and VAU-45(V)3/A. The press-to-test button shall enable manual simulation of acceleration forces.

3.9.3 Direction of acceleration force. The valve shall be marked with an arrow in accordance with Figure 1, 2 or 3 denoting the direction the influencing acceleration force shall be applied.

3.9.10 Workmanship. The valves shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance shall not authorize submission of known defective material, either indicated or actual, nor shall it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Test conditions

4.3.1 Environmental conditions. Unless otherwise specified, all tests shall be conducted at local ambient temperature and barometric pressure. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions shall be 29.92 inches of mercury and 70°F.

4.3.2 Acceleration force. When conducting the tests specified herein, the acceleration force (see 6.4) shall be applied to the valve in the direction specified on Figures 1, 2 or 3. Weights or any other means may be used to simulate the acceleration force provided the weight of the sensing mass of the valve shall be controlled, and verified by 100 percent inspection. In this case the total G force shall be 1 Gz plus the amount simulated by the test weights. The acquisition activity reserves the option to verify the performance of the valve on a centrifuge as specified in 4.6.12.

4.3.3 Test medium. Unless otherwise specified, the test medium used in testing the valves shall be air or nitrogen. The moisture content of the air shall not exceed 0.02 milligram per liter. The nitrogen shall conform to type I, class 1, grade B of BB-N-411. When nitrogen is used as a test medium, density corrections shall be made in determining that the test requirements are met.

4.3.4 Test equipment

4.3.4.1 Tank. When specified, a rigid tank having a volume of 10 $\pm$0.05 liters shall be used. The tank shall be connected to the outlet port of the valve as specified in 4.3.4.2 and shall provide a connection for the measurement of pressure in the tank. Unless otherwise specified, the sensing line port (if applicable) shall be capped for all tests.

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4.3.4.2 Test set-up. The tank specified in 4.3.4.1 shall be connected to the valve by piping which simulates the resistance characteristics of the anti-G suit hose, pressure sense line (if applicable), and applicable aircraft quick disconnect fittings. As a minimum, inlet and outlet pressure and the applied G force (if applicable) shall be measured. Figure 5 can be used as a guide. The pressure sensors shall have a flat frequency response from 0.0 to 1000 Hz.

4.4 First article inspection. First article inspection shall be as specified by the acquisition activity. Table III lists of all the examinations and tests applicable to this specification.

4.4.1 First article samples. The samples shall be of identical construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these valves, from contract to contract, first article inspection on the new contract may be waived at the discretion of the acquisition activity (see 6.3). Approval of the first article inspection samples or the waiving of the first article inspection shall not preclude the requirements for performing the quality conformance inspection.

4.4.2 Disposition of sample units. Upon completion of the first article inspection, all the applicable inspection reports, sample valves, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant Government activity. One of the approved first article inspection samples of the valves shall be returned to the contractor for use in monitoring production. The other valves shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract or purchase order.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table IV. The sampling and inspection levels and acceptance criteria shall conform to MIL-STD-105. The lot from which the valve samples were selected shall be rejected if any sample fails to comply with any requirement specified. A rejected lot shall not be resubmitted except with the approval of the contracting officer.

4.5.1 Endurance acceptance criteria. The endurance requirements of 3.6.12 shall be verified by lot sampling tests to be conducted at the contractor's facility in accordance with the criteria specified in 4.6.13. The acceptance criteria for the first sampling test shall be 30,060 cycles with zero failures. The 30,060 cycles shall be accumulated on two units, 15,030 cycles on each unit, selected from the first 10 units produced under this specification. One unit out of every subsequent production lot of 500 shall be subjected to 15,030 cycles to the criteria stated above with zero failures as the acceptance criteria. Corrective actions for all failures shall be determined and recommended to the acquisition activity. In the event of non-compliance (reject decision), or when a pattern failure occurs, the contractor shall immediately notify the acquisition activity. Changes of specified performance or required endurance characteristics of the valves are not corrective action.

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Table III. First article inspection.

Requirement Inspection	Examination or paragraph	Examination or test method
Visual examination		4.6.1
Minimum operating acceleration force	3.6.1	4.6.2
Outlet pressure regulation	3.6.2	4.6.3
Leakage	3.6.3	4.6.4
Anti-chatter	3.6.4	4.6.5
Response time	3.6.5.2	4.6.6
High temperature exposure	3.6.6	4.6.7
Low temperature exposure	3.6.7	4.6.8
Vibration	3.6.8	4.6.9
Salt fog 1/	3.6.9	4.6.10
Internal corrosion 1/	3.6.10	4.6.11
Centrifuge testing	3.6.11	4.6.12
Endurance 1/	3.6.12	4.6.13
Disassembly and examination	3.6.13	4.6.14

1/ This test shall be considered destructive. Valves subjected to this test shall not be considered or shipped as part of the contract or purchase order. This test shall be conducted on new valves ready for delivery to the Government.

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TABLE IV. Quality conformance inspection.

Test or inspection	Test method	Sample size	Classification of characteristics	Acceptance criteria
Visual examination. Inspect for relief valve safety wired	4.6.1	Every valve for critical defects Inspection level II for minor defects	Critical Minor	Reject all valves with any critical defects. An acceptable level of 2.5 defects per hundred
Dimensions and Configuration	4.6.1.1.1	Inspection level S-2	Major	Acceptance number of 0, rejection number of 1.
Minimum Operating Acceleration Force	4.6.2	Every valve	Major	Acceptance number of 0, rejection number of 1.

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TABLE IV. Quality conformance inspection. - Cont'd

Test or inspection	Test method	Sample size	Classification of characteristics	Acceptance criteria
Outlet Pressure Regulation	4.6.3	Every valve	Critical	Acceptance number of 0, rejection number of 1.
Leakage	4.6.4	Every valve	Major	Reject all defective units.
Anti-Chatter	4.6.5	Inspection level I	Major	Acceptance number of 0, rejection number of 1.
Response Time	4.6.6	Every valve	Critical	Reject all defective units.
Endurance 1/	4.6.13	See 4.5.1	Major	See 4.5.1
Packaging	4.6.1.2	Inspection level S-2	Minor	Total acceptable quality level of 4.0 percent defective.

1/ This test shall be considered destructive. Valves subjected to this test shall not be considered or shipped as part of the contract or purchase order. This test shall be conducted on new valves ready for delivery to the Government.

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4.5.2 Sampling

4.5.2.1 Inspection lot

4.5.2.1.1 Valve. An inspection lot size shall be expressed in units of valves, manufactured essentially under the same conditions and from the same materials and components. The sample unit shall be one valve.

4.5.2.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing valves fully prepared for delivery and made from essentially the same materials and components. The sample unit shall be one shipping container, containing valves fully prepared for delivery with the exception that it need not be sealed.

4.5.2.2 Sampling for tests and inspections of valves and packaging. The sample size, acceptance criteria, tests, and inspections required for the valves and for the packaging shall be as specified in Table IV.

4.6 Inspection methods

4.6.1 Visual examination

4.6.1.1 Valve. Each valve selected as a sample unit from the lot shall be thoroughly examined visually and checked dimensionally for defects to determine conformance to this specification with respect to materials, design, construction, marking, and workmanship. Defects shall be classified in accordance with Table V.

4.6.1.1.1 Dimensions and configuration. The valves shall be examined for defects in dimensions and configuration. Any dimensions or configuration not conforming to the requirements of the applicable assembly number of Figures 1, 2 or 3, as applicable, shall be classified as a major defect.

4.6.1.2 Packaging. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with Table VI. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundreds units, shall be 2.5.

4.6.2 Minimum operating acceleration force. At the start of this test the press-to-test button shall be depressed to ensure air flow at the outlet. No air flow at the outlet with the button fully depressed shall be cause for failure of this test. The valve shall be subjected to inlet pressures of 30, 70, 150, and 300 psig. The acceleration force required to open the valve at each inlet pressure shall be as specified in 3.6.1.

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TABLE V. Classification of defects for visual examination of the valve.

Critical	Minor
1. Material imperfections – foreign matter embedded.	1. Marking – missing, insufficient, incorrect, permanent.
2. Surface – unclean, rough, misaligned, or containing cracks, nicks, or other flaws.	
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	
8. Linkage interference or component binding.	

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Table VI. Classification of defects for packaging examination.

Examine	Defect
Marking (exterior interior)	Missing, incorrect, incomplete, and illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any nonconforming component;any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components, such as: incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc. Bulging or distortion of the containers.
Exterior and interior	Number per container is more or less than content and weight required; gross or net weight exceeds the requirements.

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4.6.3 Outlet pressure regulation. A tank shall be connected as specified in 4.3.4 during this test. A source of pressurized gas shall be connected to the valve inlet port. The inlet gas pressure shall be increased and maintained at 30 psig while equivalent acceleration forces of 2, 4, 6, 8, 9, 10, 9, 8, 6, 4, 2, and 1 Gz shall be applied in this order to the valve. Each force shall be maintained for at least 10 seconds before the next force shall be applied. The inlet pressure shall be increased to 70, 150 and 300 psig, and the test repeated at each of these inlet pressures. The outlet pressure shall be as specified in 3.6.2.

4.6.4 Leakage. With a force of 1 Gz applied to the valve, the inlet flow to the valve at an inlet pressure of 30, 70, 150, and 300 psig shall not exceed that specified in 3.6.3

4.6.5 Anti-chatter. A tank shall be connected as specified in 4.3.4 during this test. A source of pressurized gas shall be connected to the valve inlet connection. The inlet pressure shall be increased and maintained at 30, 70, 150 and 300 psig. At each pressure level the valve shall be fully opened manually, by pressing the press-to-test button, and held open to determine that there is no evidence of chatter when there is air flow through the relief valve. Additional anti-chatter testing shall be as detailed in 4.6.12.2.5.

4.6.6 Response time. A tank shall be connected as specified in 4.3.4 during this test. A source of pressurized gas shall be connected to the valve inlet connection and the pressure increased to and maintained at 70 psig. A pressure transducer shall be connected to the tank to allow tank pressure to be recorded as a function of time. Acceleration forces of 6, 1, 10 and 1 Gz shall be applied in this order as a step input to the valve. Each force shall be maintained for at least 10 seconds before the next force shall be applied. Tank pressure shall be recorded as a function of time for at least 5 seconds following the application of each force level. The time interval from the time of application of each force level to the time at which the tank pressure reaches the value specified in 3.6.5.2 shall not exceed the response time specified in 3.6.5.2.

4.6.7 High temperature exposure. The pressure regulating valve shall be conditioned at a temperature of 250°F ±5°F for eight hours. After this period, the valve shall be immediately subjected to and pass the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5, and 4.6.6; except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.8 Low temperature exposure. The pressure regulating valve shall be stabilized at a temperature of -65 ±5°F. After this period, the valve shall be warmed to -40 ±5 °F and immediately subjected to and pass the tests specified in 4.6.2, 4.6.3, 4.6.4, 4.6.5 and 4.6.6; except that the minimum operating force can be between 1.25 and 2.0 Gz.

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4.6.9 Vibration

4.6.9.1 Random vibration. With the valve connected to the test equipment described in 4.3.4, with an inlet pressure of 300 psig, the valve shall be exposed for one hour at each simulated G level to the random vibration described below in each of three mutually orthogonal axes. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.6.2 through 4.6.6, except that the minimum operating force can be between 1.25 and 2.0 Gz. The random vibration envelope (see Figure 6) shall be as follows:

0.04 g^2/Hz from 15 - 100 Hz
-4 db/octave slope from 100 - 140Hz
0.025 g^2/Hz from 140 - 540 Hz
0.072 g^2/Hz from 540 - 900 Hz
-6 db/octave from 900 - 2000 Hz

4.6.9.2 Resonance. With the test equipment and anti-g valve set up as described in 4.6.9.1, and using a sinusoidal vibration at 0.02 inches or ±2 Gs double amplitude whichever shall be less; a resonance search shall be conducted between 5 and 2000 Hz in three mutually orthogonal axes to identify resonant frequencies. The valve shall then be subjected to any and all resonant frequencies for seven and one half minutes each. There shall be no damage to the valve and the valve shall meet the requirements of sections 3.6.1 through 3.6.5 before and after this exposure. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.6.2 through 4.6.6, except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.9.3 Gunfire Vibration. With the test equipment and anti-g valve operating as described in 4.6.9.1, the valve shall be subjected to the following vibrations along each of three mutually orthogonal axes. After this test, there shall be no damage to the valve and the valve performance shall be verified as described by 4.6.2 through 4.6.6, except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.9.3.1 Excitation. The excitation shall consist of four phase uncorrelated sinusoids superimposed over a random vibration background. The sinusoids shall be at frequencies representing the basic gunfire rate orders; 100 Hz (f_1), 200 Hz (f_2), 300 Hz (f_3), and 400 Hz (f_4). Synchronously, the frequencies shall be swept logarithmically over a bandwidth equal to ±0.2 times f_1, f_2, f_3, and f_4, beginning at the low frequency and ending at the high frequency limit. The sweep time for the combination sine sweep and random vibration shall be 60 minutes per axis. The instantaneous peaks of the random vibration may be limited to 2.5 times the root mean square spectral acceleration level. Spectrum frequencies, frequency-function levels, and location-function levels for the combined random and sinusoidal vibration tests shall be as contained in Figure 7. After this test, there shall be no damage to the valve and the valve performance shall be verified as described in 4.6.2 through 4.6.6, except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.9.3.2 Resonance dwell. The anti-g valve shall be subjected to eight resonance dwells per orthogonal axis, each lasting five minutes. The excitation level at the dwell

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frequencies shall be as described below and shown in Figure 7. Background random excitation shall not be required for the following gunfire resonance tests. Upon completion of these tests, the valve performance shall be verified as described in 4.6.2 through 4.6.6, except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.9.3.2.1 Frequency band. One dwell shall be done in each of the frequency bands corresponding to ±10 percent of each of the above gunfiring frequencies (f_1, f_2, f_3, and f_4). If no resonance is found in a gunfire frequency band, the dwell shall be performed at the center frequency of the band.

4.6.9.3.2.2 Dwells. Dwells shall be conducted in the frequency range from 100 to 1000 Hz at the four most significant resonant frequencies from section 4.6.9.2.

4.6.10 Salt fog. The valve shall be subjected to a salt fog test in accordance with method 509.2 of MIL-STD-810. The valve shall be examined for evidence of corrosion to all exposed surfaces. If the valve exhibits any corrosion indicative of dissimilar metals or corrosion that could preclude normal operation or create any unsafe condition, the valve shall be rejected. The valves shall be subjected to and pass the tests described in 4.6.2 through 4.6.6; except that the minimum operating force can be between 1.25 and 2.0 Gz and the outlet pressure can vary by an additional ±1.0 psi.

4.6.11 Internal corrosion. With the valve maintained at a temperature of 135 ±5°F, a salt fog in accordance with the salinity requirements of MIL-STD-810, method 509.2, at a temperature of 135 ±5°F, shall be passed through the valve in the normal direction of flow at approximately three flow cycles per minute for a duration of at least 15 minutes. The rate of flow of salt fog shall be no less than 30 liters per minute. This procedure shall be performed at least 10 times with the valve dried after each 15 minute period. After completion of the above tests, the valve shall be subjected to and pass the tests specified in 4.6.2 through 4.6.6; except that the minimum operating force can be between 1.25 and 2.0 Gz and the outlet pressure can vary by an additional ±1.0 psi.

4.6.12 Centrifuge testing. Centrifuge testing shall be conducted at a facility approved by the Government. The following shall be capable of being recorded.

a. Gz: 1 to 10 G with an accuracy of ±0.1 G

b. Tank or anti-g suit pressure: 0 to 15 psig, with an accuracy of ±0.1 psig

c. Valve outlet pressure: 0 to 15 psig, with an accuracy of ±0.1 psi

d. Source pressure: 0 to 300 psig, with an accuracy of ±10 psig

4.6.12.1 Sampling rates and calibrations. The sampling rates and calibrations shall ensure compliance with the performance requirements of this specification. Both pre-test and post-test calibrations shall be required. The valve shall be mounted in radial line with the accelerometer, and connected in accordance with 4.3.4.2 to a tank as specified in 4.3.4.1. The test medium shall be as specified in 4.3.3 and the test set-up shall be in accordance with 4.3.4.2, with the exception of 4.6.12.2.5 which includes instructions for set-up.

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4.6.12.2 Tests. The following tests shall be performed:

4.6.12.2.1 Test profile 1. With the valve inlet pressure maintained at 30 psig, the centrifuge shall be accelerated from 1.2 Gz to 10 Gz, held at 10 Gz for at least 10 seconds, and then decelerated back to 1.2 Gz. The acceleration and deceleration (onset/offset) rates shall be 0.1 Gz/sec. This shall be as shown in Figure 8. Gz level at pressure initiation, the valve outlet pressure versus Gz level, and the relief valve opening point shall be in accordance with section 3.5.1 and 3.6.2 of this specification. The test shall then be repeated for a 4 Gz/sec onset/offset rate. The above test, at both onset/offset rates, shall be repeated for 70, 150 and 300 psig valve inlet pressures.

4.6.12.2.2 Test profile 2. The valve inlet pressure shall be maintained at 30 psig while the centrifuge is incrementally accelerated from 1.2 Gz to Gz levels of 2, 4, 6, 8 and 10 at an onset rate of 4 G/sec. Each G level shall be maintained for 10 seconds. It shall then be incrementally decelerated to 8, 6, 4, 2, and 1.2 Gz at a rate of 4 G/sec with 10 seconds at each level. This shall be as shown in Figure 9. The valve outlet pressure shall be in accordance with sections 3.6.2 through 3.6.5 of this specification. The above tests shall then be repeated with inlet pressures of 70, 150 and 300 psig.

4.6.12.2.3 Rotated mounting. Mount the test valve so that the direction of valve's acceleration sensor shall be 22 degrees from the direction of the applied acceleration, in the direction of the applied acceleration, in the direction of the relief valve (see Figure 10). With the inlet pressure at 70 psig, accelerate the test valve at an onset rate of 0.1 G/sec, to 6 G's, hold at this level for 10 sec, and decelerate at an offset rate of 0.1 G/sec to normal gravity. The valve outlet pressure, with respect to the acceleration in the direction of the acceleration sensor, shall be as specified in Section 3.6.2. The valve acceleration shall be determined by multiplying the actual acceleration by Cosine (22°).

4.6.12.2.4 Repeat test. Repeat the tests of 4.6.12.2.3 using an onset and offset rate of 4 G/sec and ensuring that the valve's performance meets the requirements of Section 3.6.2

4.6.12.2.5 Test for anti-chatter. The valve shall be checked for anti-chatter by mounting an MXU-804/A or an MXU-805/A in the centrifuge and connecting it to a standard anti-g suit. A profile that shall cause this valve to chatter shall be run (approximately 3-5 Gz with an onset rate of 3 G/sec; the inlet to the anti-g suit may need to be restricted to induce the chatter). After chatter has been induced in this valve, replace the valve with the valve under test without disturbing the rest of the set-up. Run the same profile that caused the chatter and insure no chatter shall be present in the valve under test.

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4.6.13 Endurance. A tank shall be connected as specified in 4.3.4 during this test. A source of pressurized gas shall be connected to the inlet and the valve shall be subjected to cyclic operations for the number of cycles (see 6.4.3), under the conditions, and in the order specified in Table VII. The rate of cycling shall be approximately 6 cycles per minute. The outlet pressure at the maximum force of each cycle shall be in accordance with 3.6.2 for the applied force. After completing all phases of the cyclic operation, the valve shall be subjected to and pass the inspections specified in 4.6.2 through 4.6.6; except that the minimum operating force can be between 1.25 and 2.0 Gz.

4.6.14 Disassembly and examination. At the conclusion of all inspections specified herein, the valve shall be disassembled and examined. The valve shall not show evidence of excessive deterioration or wear due to testing which might affect the valve performance. The valve shall be disassembled and examined after any test in which valve failure indicates faulty construction.

5. PACKAGING

5.1 Preservation-Packaging. Preservation-Packaging shall be level A in accordance with MIL-STD-2073-1. Preservation and packaging shall be accomplished in accordance with FED-STD-209.

5.1.1 Level A. Unless otherwise specified, valves, pressure, anti-G suit, VAU-45(V)1/A, VAU-45(V)2/A and/or VAU-45(V)3/A shall be preserved in accordance with method ICI of MIL-P-116.

5.1.1.1 Cleaning and drying. Each valve shall be cleaned and dried for oxygen service in accordance with industry practices, according to the applicable processes of MIL-P-116. Petroleum and other flammable solvents shall not be used.

5.1.1.2 Preservation application. No preservatives shall be applied.

5.2 Packing

5.2.1 Unit packaging. Unless otherwise specified by the acquisition activity, valves shall be individually packaged in accordance with MIL-P-116, Method ICI. Unit packaging shall meet all applicable requirements of MIL-P-116.

5.2.2 Level B. Unless otherwise specified by the acquisition activity, valves preserved as specified in 5.1, shall be packed in shipping containers conforming to PPP-B-636, style RSC, type CF, Class WR, grade V3c. Additional requirements as specified in 5.2.1 shall be applicable.

5.2.3 Level C. Unless otherwise specified by the acquisition activity, valves preserved as specified in 5.1, shall be packed in shipping containers conforming to PPP-B-636, style RSC, type CF, class DOM, grade 275. Other requirements as specified in 5.2.1 shall be applicable.

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TABLE VII. Conditions of cyclic operation.

Inlet pressure (psig)	Inlet air temperature (°F)	Applied force (Gz)	Number of cycles
150	70	1 to 6 to 1	5,000
150	70	1 to 9 to 1	15
300	400	1 to 6 to 1	1,000
300	400	1 to 9 to 1	15
150	400	1 to 6 to 1	4,000
150	70	1 to 6 to 1	5,000

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5.3 Marking. In addition to any other markings required by the contract or purchase order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each package:

LIFE SUPPORT ITEM

All oil, grease, shop residue
or other contaminants
have been removed.

DO NOT OPEN UNTIL READY FOR USE

5.3.2 Bar code markings. Bar code markings, when specified (see 6.2), shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pressure-regulating valves are intended for use in high performance aircraft. Valves designated VAU-45(V)3/A are to be used in the F-15 aircraft with or without PBG capability, valves designated VAU-45(V)2/A are to be used in the F-16 aircraft having PBG capability, and valves designated VAU-45(V)1/A are to be used in A-10 aircraft which are not going to have PBG capability. The VAU-45(V)1/A can also be used in F-16s which do not yet have PBG capability. The valves sense change in acceleration force and provide pressure for the operation of aircrew members' anti-G suits and g-compensated oxygen regulator (if applicable) in accordance with these changes.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Part number required (see 3.9).

d. Requirements for first article samples (including shipping instructions) or waiver or requirements for first article samples (see 3.1, 4.4, and 6.3).

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e. Selection of applicable levels of preservation and packing required (see 5.1 and 5.2).

f. When bar code markings are required (see 5.3).

g. Requirements for test procedures.

6.3 First article. When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.4.1. The first article should consist of four valves. Two valves shall be subjected to the verifications of 4.6.1 through 4.6.11. The other two valves shall be subjected to the verifications of 4.6.1 through 4.6.6 and 4.6.13. All valves shall be subjected to 4.6.14 as the final verification. At the discretion of the acquisition activity, the centrifuge tests of section 4.6.12 may be done using four valves of identical design and construction. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions

6.4.1 Acceleration force. An inertial force resulting from acceleration that acts upon an aircraft or person in the aircraft.

6.4.2 Chatter. Any rattling noise; vibration (or fluttering) in the outlet pressure, or vibration in the valve body or mounts. This phenomenon may be associated with delayed performance of the valve.

6.4.3 Cycle. A cycle shall consist of varying the force on the valve from 1 Gz to the maximum specified for the specific cycle and back to 1 Gz.

6.4.4 G-force. An inertial force on an object resulting from an applied acceleration due to gravity or reaction to a change of direction in unit of gravitational acceleration.

a. +Gy. Acceleration acting across a body perpendicular to its long axis in a side to side direction. Examples are a pilot in a high performance aircraft executing an uncoordinated turn and exposure to lateral buffeting.

b. +Gz. A positive acceleration acting along the z-axis of a body. Examples are aircraft recovery from a dive and turning maneuvers.

c. -Gz. A negative acceleration acting along the negative z-axis of a body axis. Examples are flying on outside loop and a nose-over.

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6.4.5 Positive breathing for G (PBG). Breathing gases are supplied under pressure to pilot's mask and at the same pressure to a vest which provides counter pressure to the chest. The pressure provided is dependent on the Gz level. A pressure sense line connects the anti-g valve to a g-compensated regulator.

6.5 Part or identifying number (PIN). The part numbering system shall be as follows:

a. Valves designated VAU-45(V)1/A shall be identified M87255-1.

b. Valves designated VAU-45(V)2/A shall be identified M87255-2.

c. Valves designated VAU-45(V)3/A shall be identified M87255-3.

6.6 Subject term (key word) listing.

Suit, Anti-G
Valve, anti-G
Valve, high flow
Valve, high flow positive breathing for G
Valve, relief
Valve, Pressure regulators

6.7 Recycled, virgin, and reclaimed materials. Provided that all other requirements of this specification are met, reclaimed materials should be considered for use whenever feasible with no exclusions to the use of recovered materials and no requirements that an item be manufactured from virgin materials.

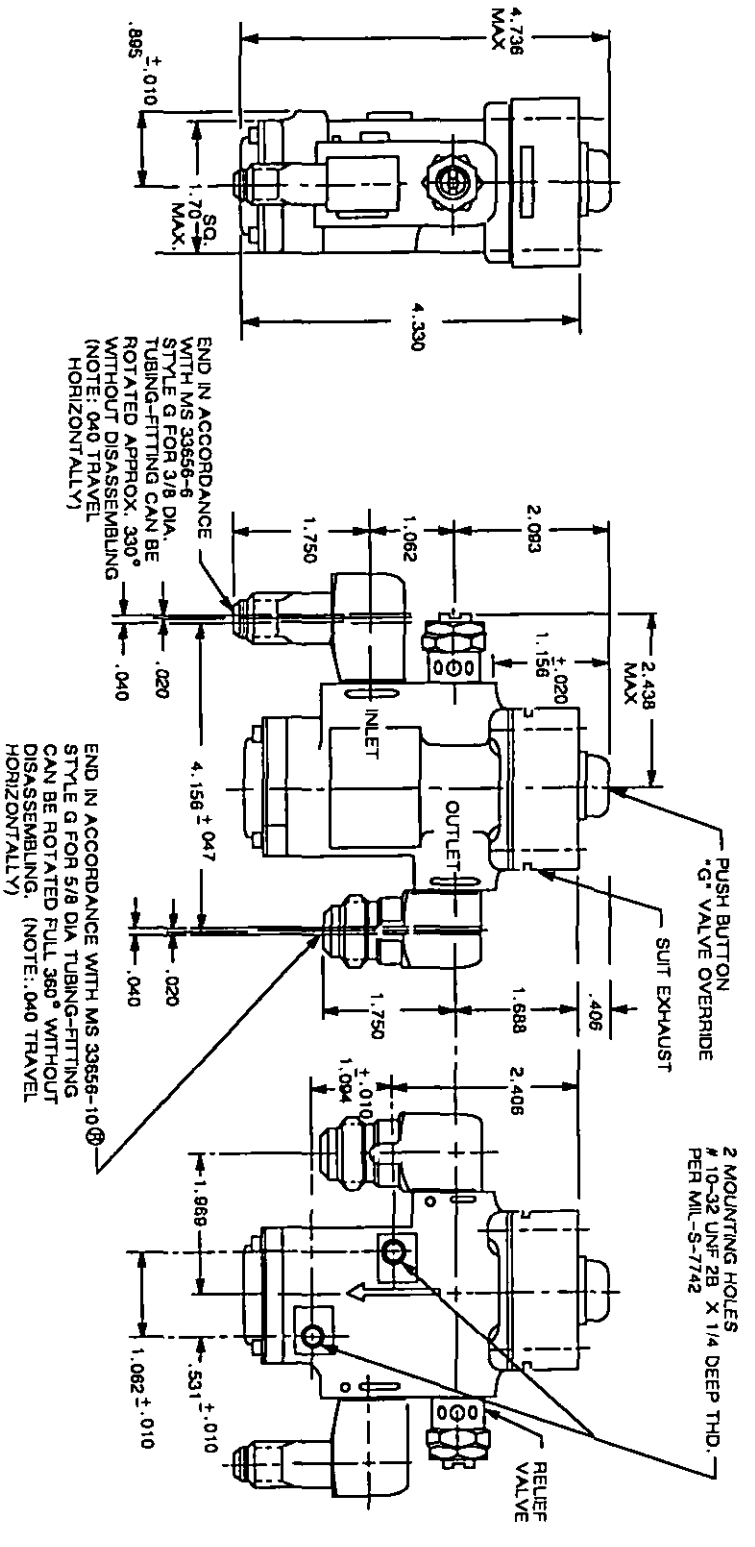
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Review activities:
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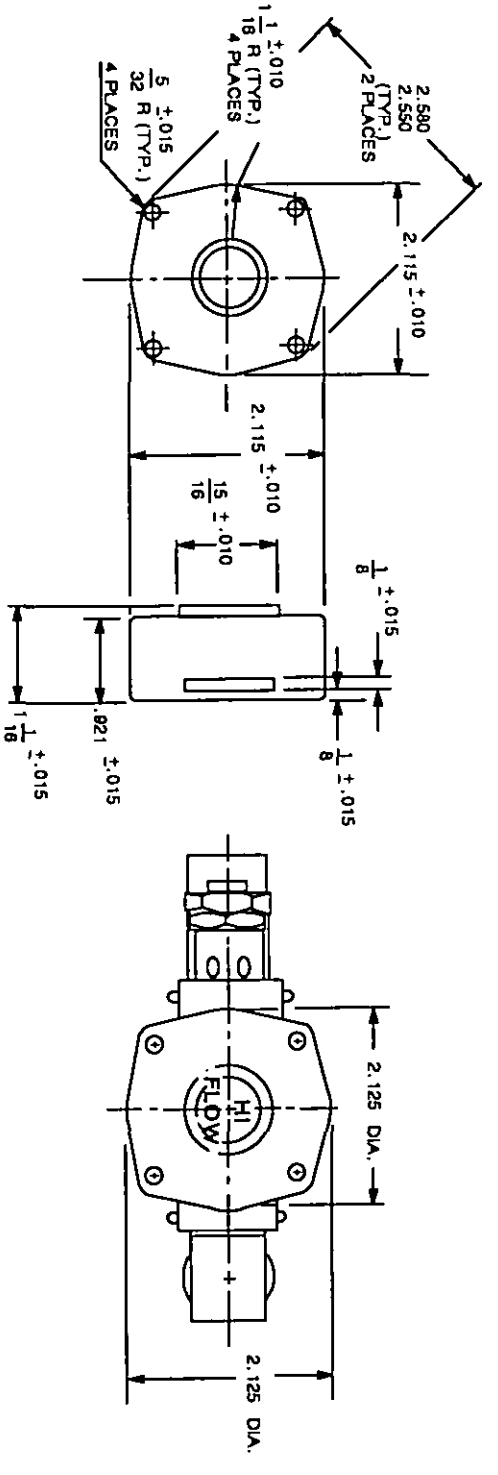
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Figure 1. Aircraft interface dimensions for VAU-45(V)1/A.

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Figure 1. Aircraft interface dimensions for VAU-45(V)1/A - continued.

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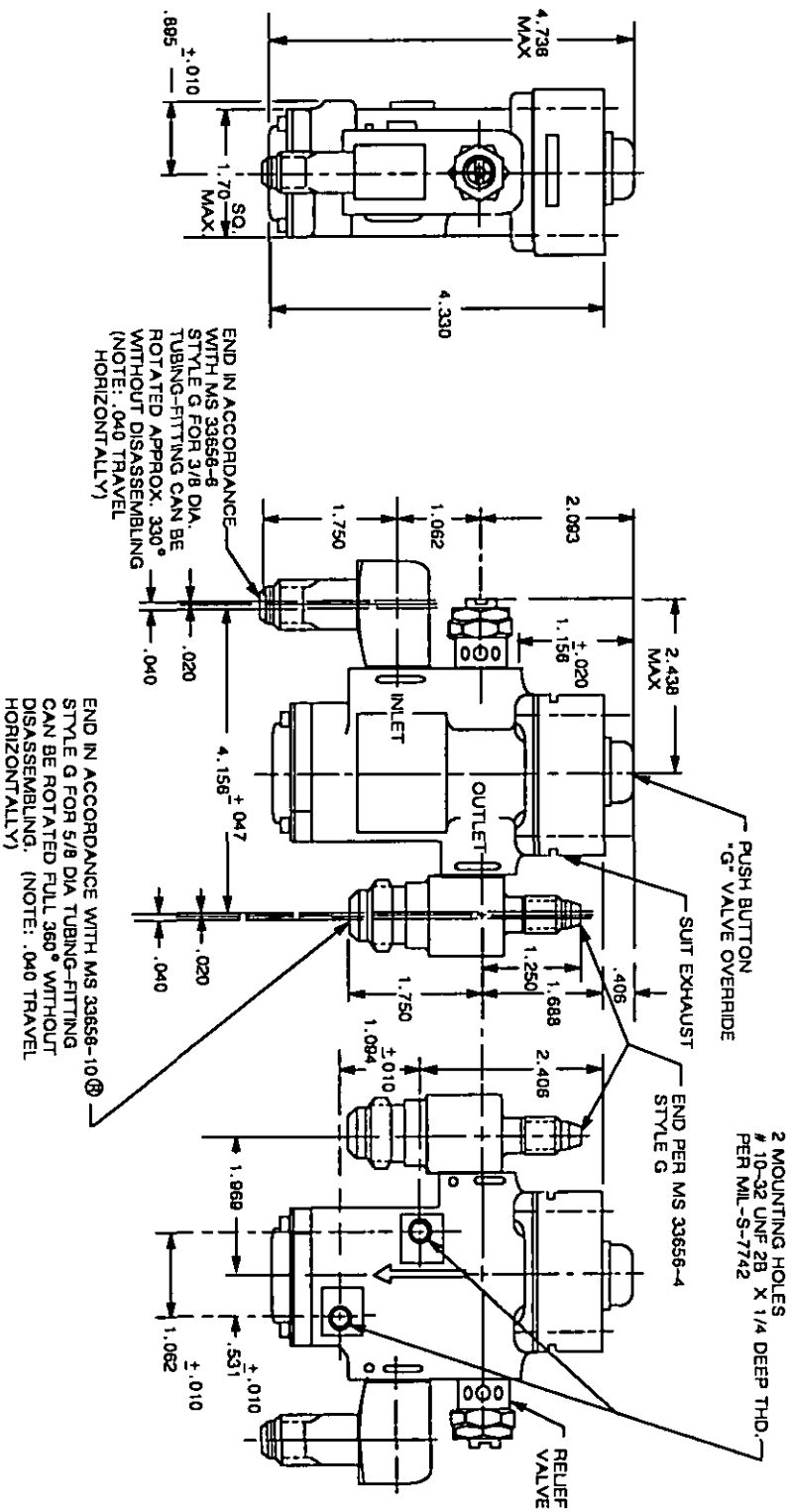


Figure 2. Aircraft interface dimensions for VAU-45(V)2/A.

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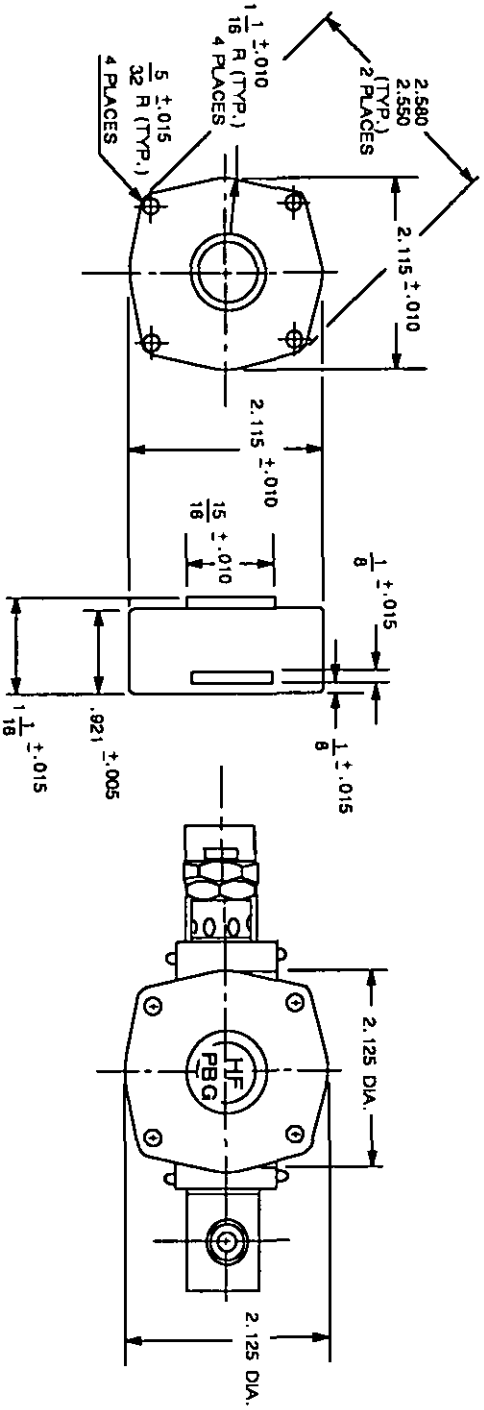


Figure 2. Aircraft interface dimensions for VAU-45(V)2/A-continued.

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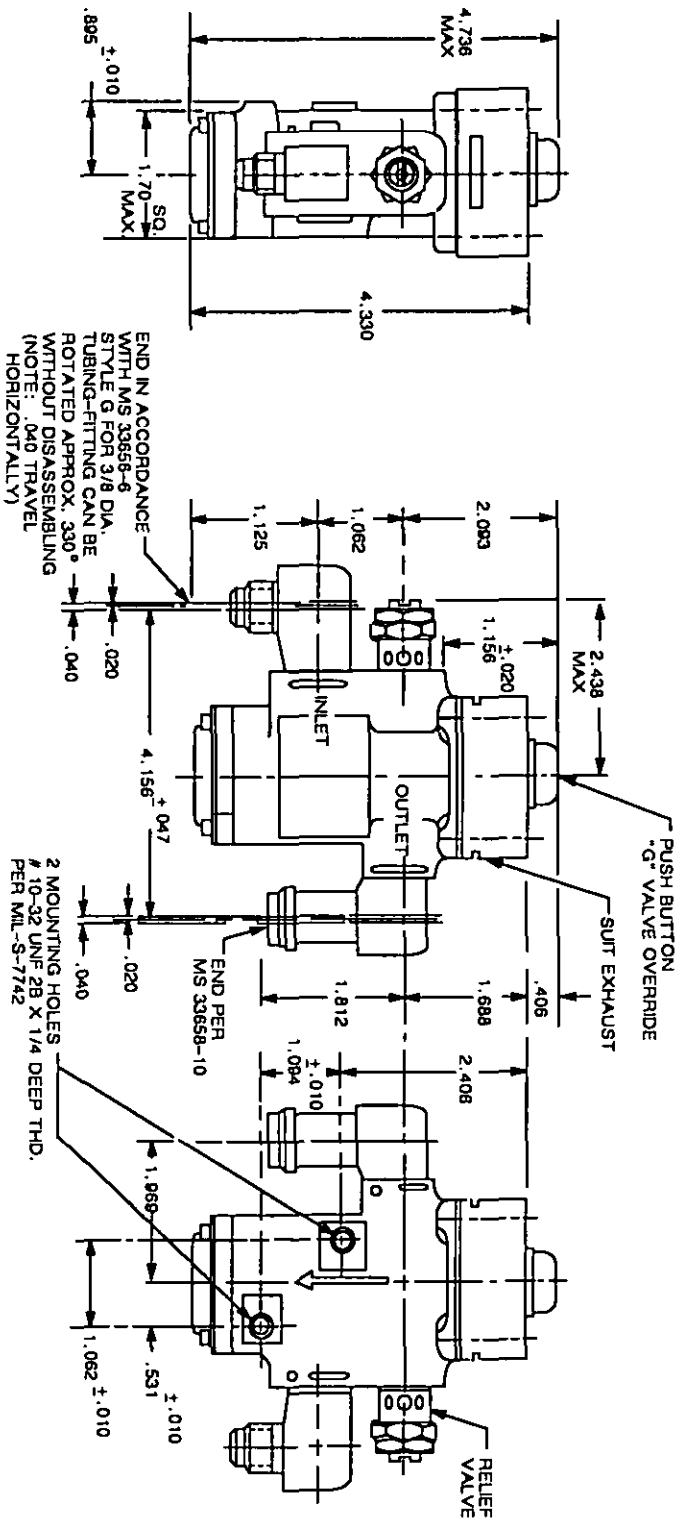


Figure 3. Aircraft interface dimensions for VAU-45(V)-3/A.

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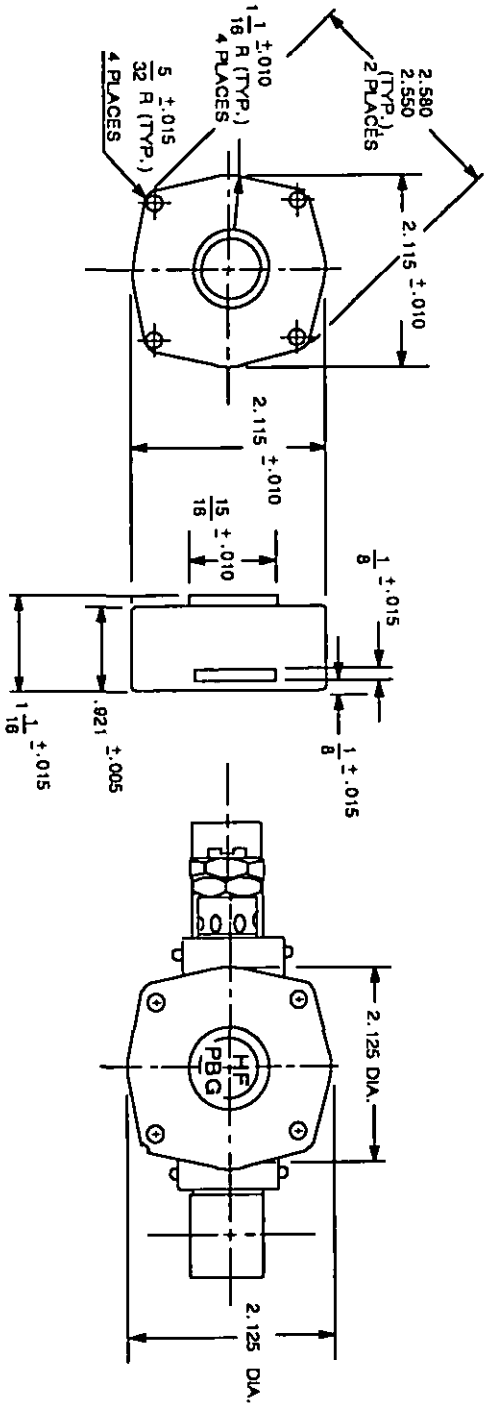


Figure 3. Aircraft interface dimensions for VAU-45(V)-3/A-continued.

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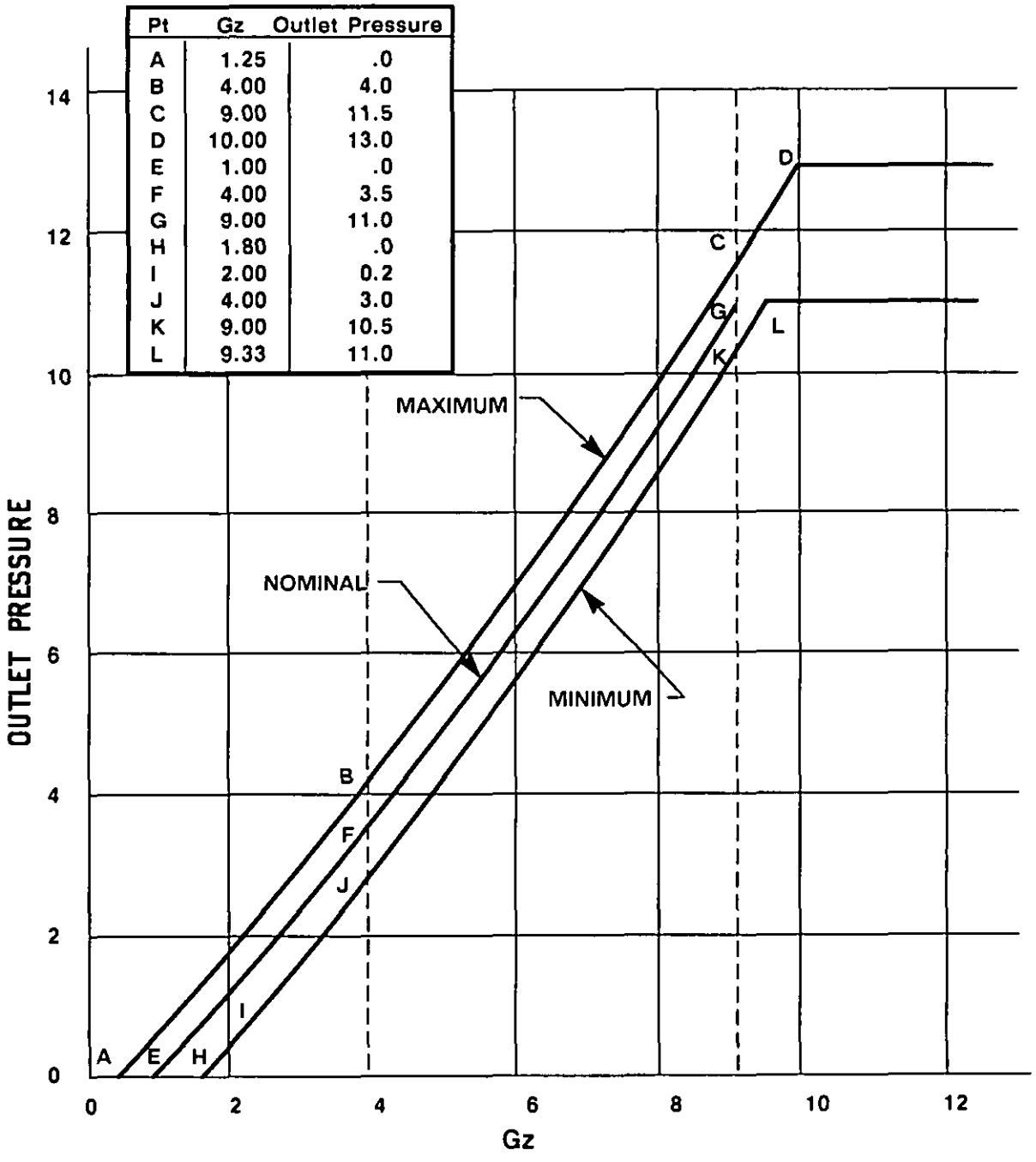


Figure 4. Anti-g valve outlet pressure vs Gz.

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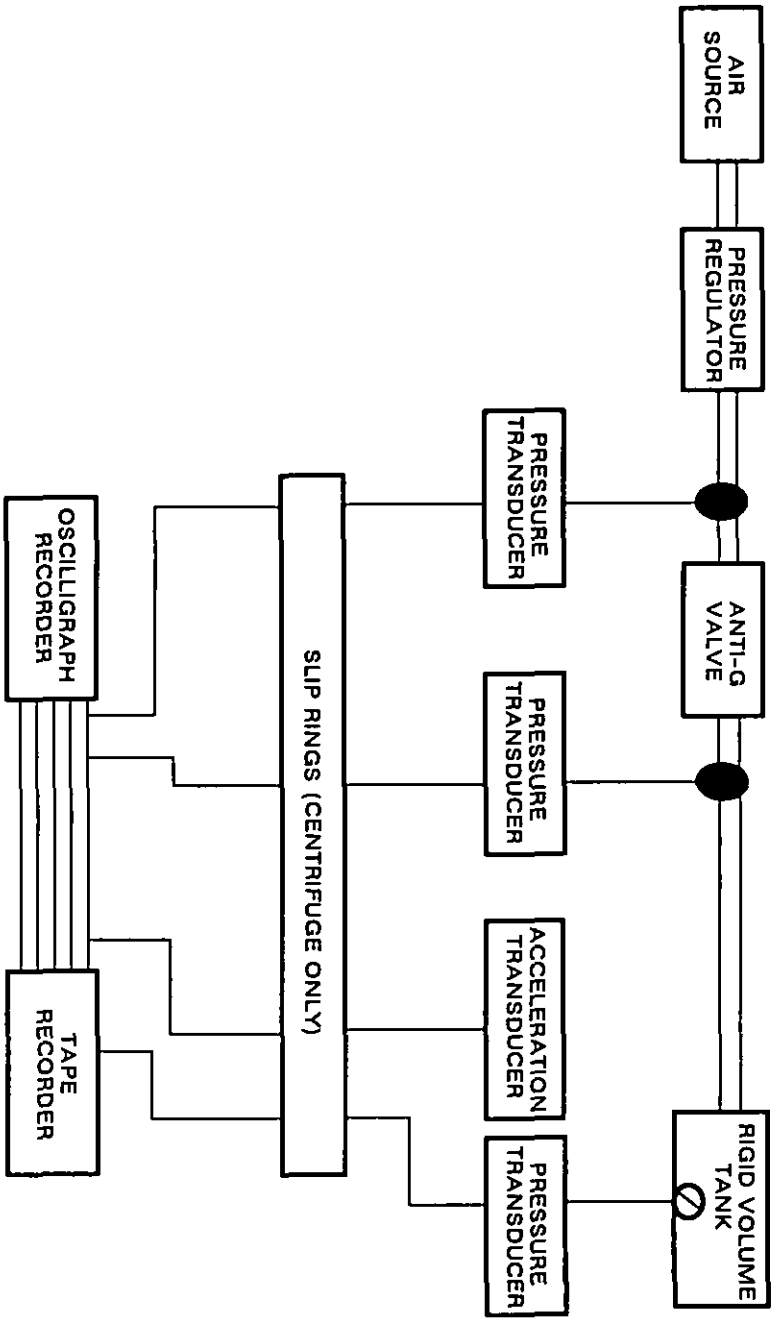


Figure 5. Test set-up.

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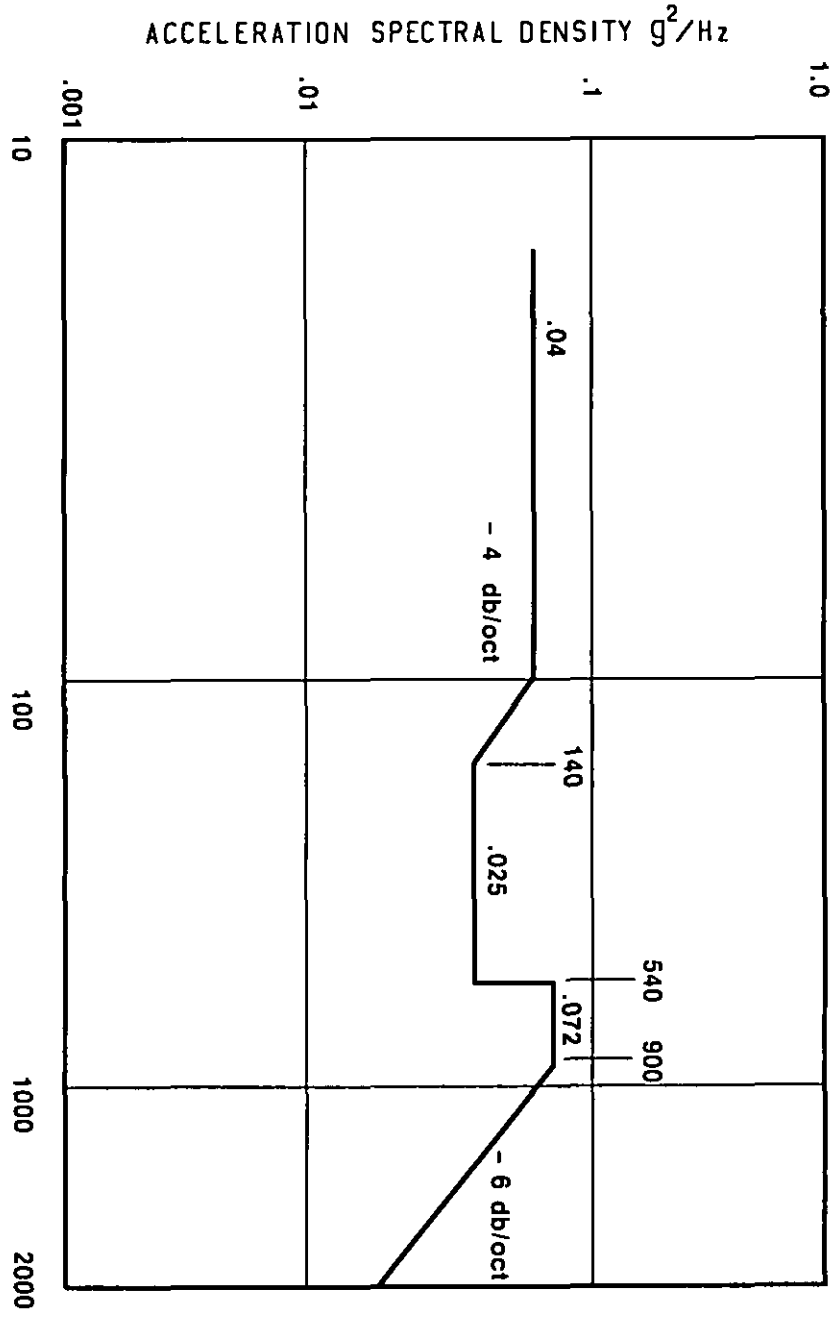


Figure 6. Random vibration versus frequency.

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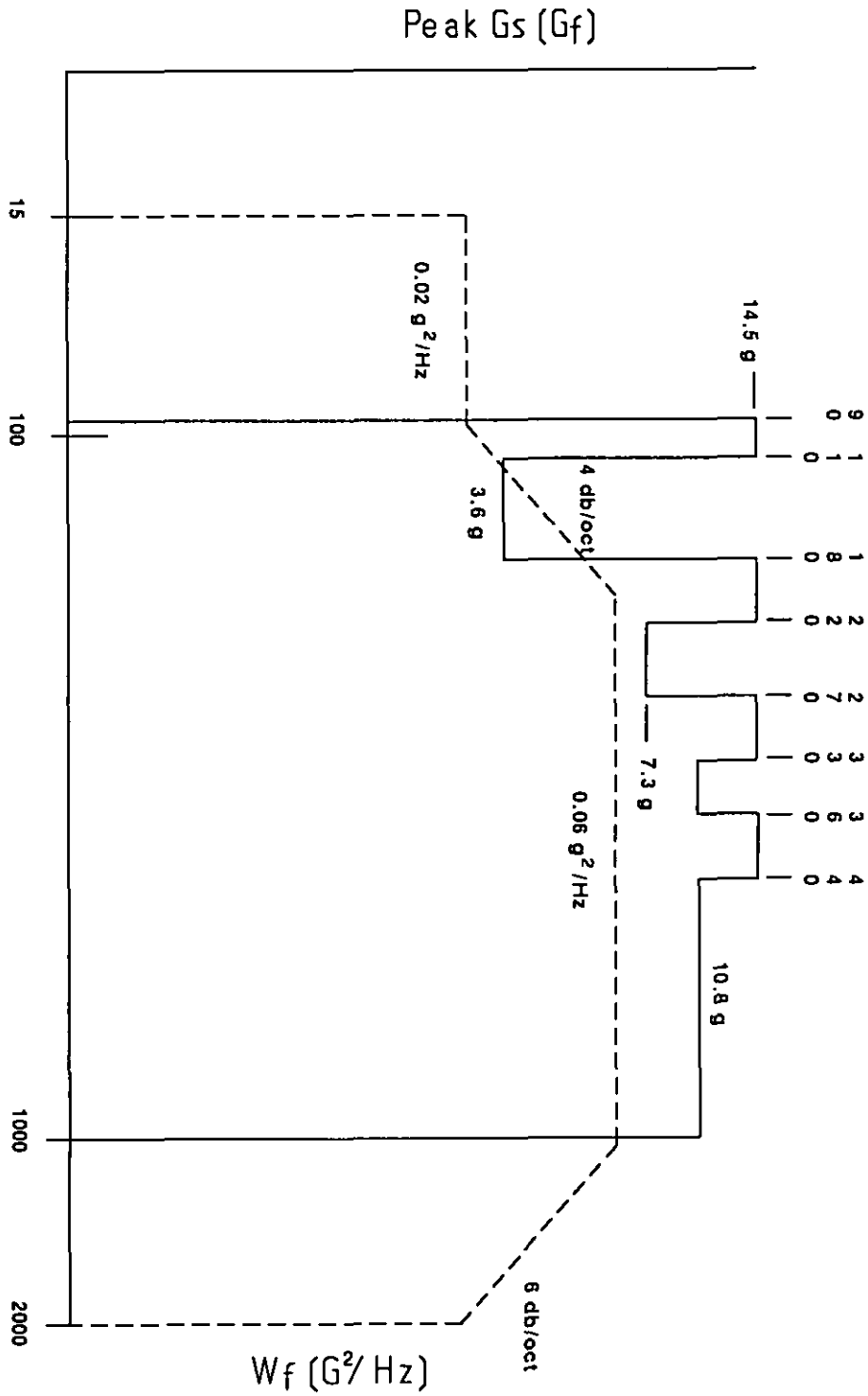


Figure 7. Combined random & sinusoidal vibration for gunfire environment.

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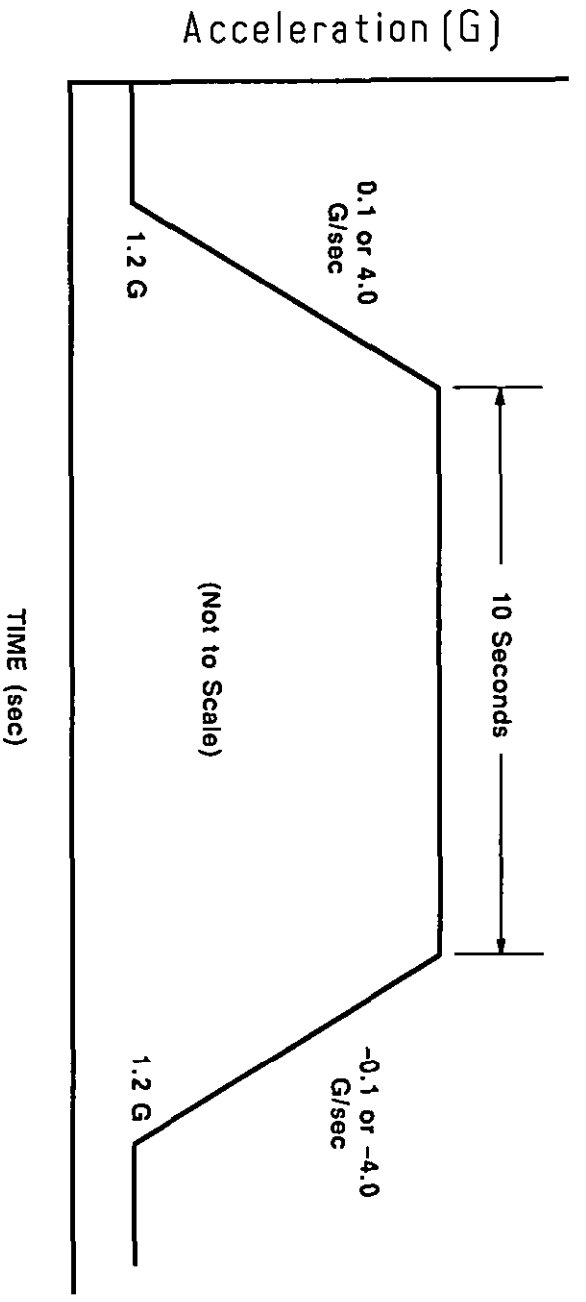
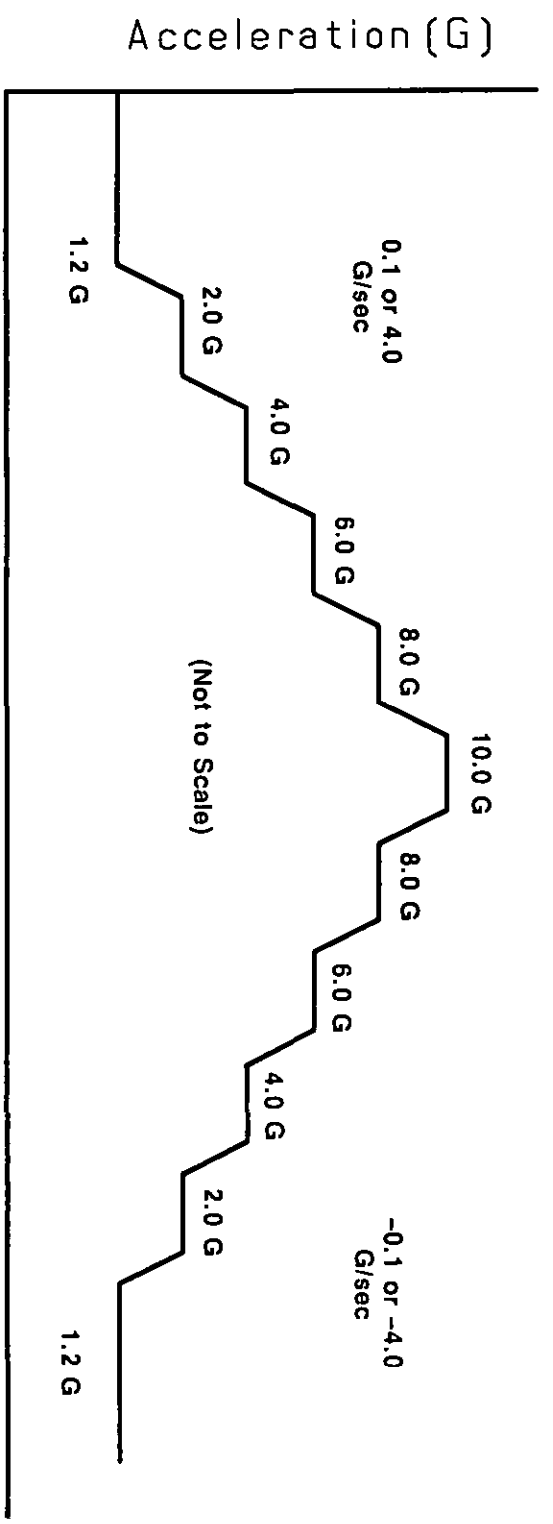


Figure 8. Test profile 1.

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Note 1. Maintain each level for 10 seconds

Note 2. Use 4.0 and -4.0 G/sec acceleration and deceleration

Figure 9. Test profile 2.

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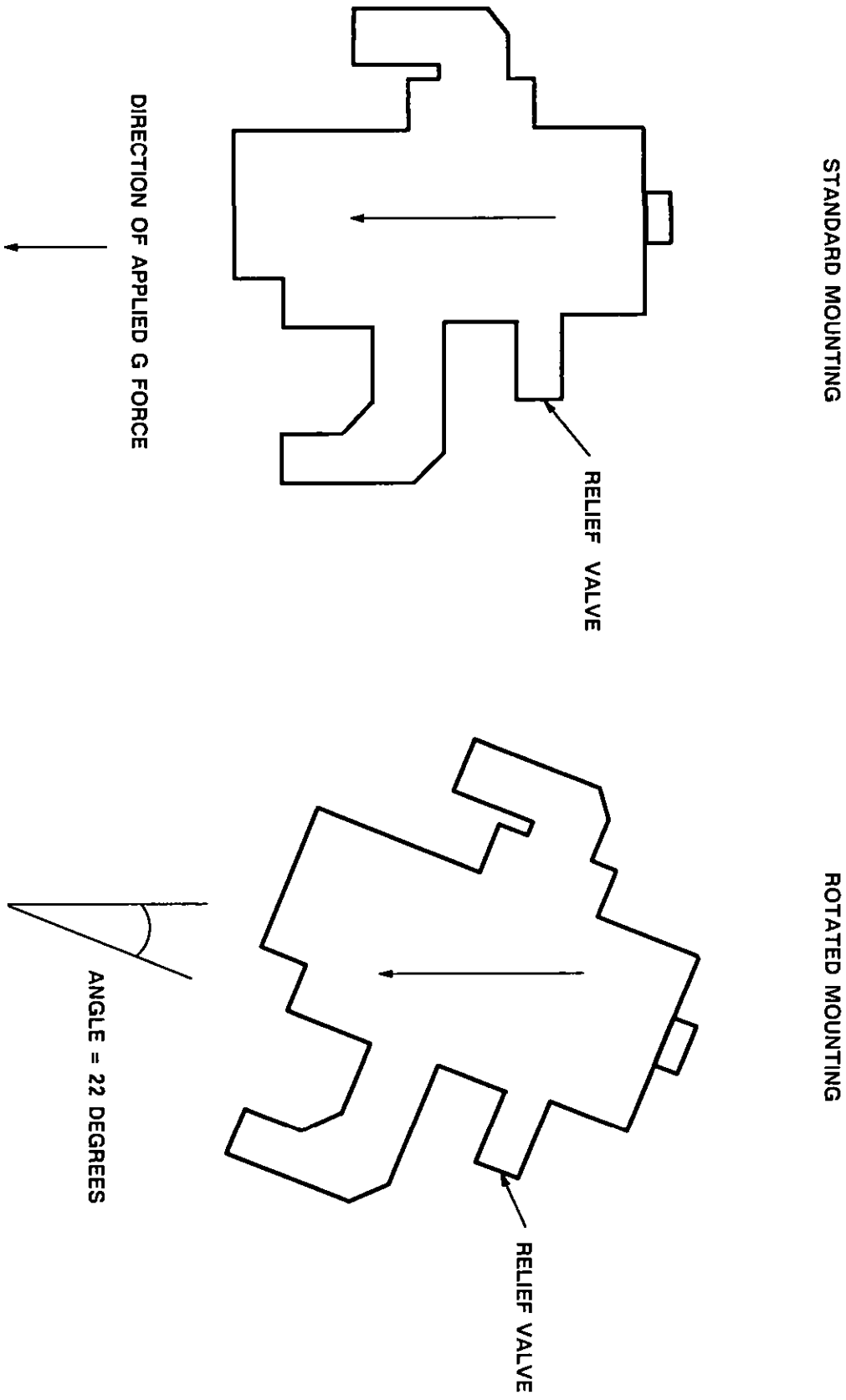


Figure 10. Rotated mounting for centrifuge testing.

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